

Constrained Graphic Layout Generation via Latent Optimization

Verified reference summary • Year: 2021 • Source bibliography: docs/REFERENCES.md

What the paper contributes

This work defines layout constraints and metrics such as alignment and overlap over element bounding boxes. It is valuable because the formulas are format-agnostic: if you know the boxes, you can measure the layout.

Method / evaluation stance

slidekit's linter and aesthetic scorer operate in precisely this regime. Overlap is exact rectangle intersection. Alignment is measured from edges and centers. No screenshot is required to know whether two boxes collide.

How slidekit uses it

The paper supports a clean separation between generation and verification. A model or component handler can propose a layout, and deterministic checks can reject or score it.

YAML vs HTML takeaway

YAML is especially well suited here because the system owns the conversion from semantic slots to boxes. HTML can expose boxes through the DOM, but CSS layout, transforms, and browser differences add another layer of uncertainty.

Bottom line

The paper supports a pragmatic split: use research-style rendered artifacts when the question depends on final pixels or human preference collection, but keep slidekit production decks in a constrained YAML IR so layout, linting, and agent fixes remain deterministic, inspectable, and cheap.

Primary slidekit mapping	balance / alignment / whitespace / overlap / contrast / richness / consistency, depending on the paper
Operational risk	HTML metrics can drift with rendering environment; YAML metrics can miss pixel-only effects. Use each where it is strongest.